Session 07 (AIML) -Assignment Question

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- **Domain Name:** AIML Development
- **College Name:** Zeal Polytechnic , Narhe
- **Branch Name:** AIML

Q1.

PROGRAM:

```
session 7.py > ...
# a) Tuple with 5 numbers
t1 = (10, 20, 30, 40, 50)

# b) Mixed tuple
t2 = (10, "Python", 3.14, True)

# c) Empty tuple
t3 = ()
t4 = tuple()

# d) Single element tuple
t5 = (99,)  # Comma is required

print(t1, type(t1))
print(t2, type(t2))
print(t3, type(t3))
print(t4, type(t4))
print(t5, type(t5))
```

OUTPUT:

```
op\ITR\session 7.py'
(10, 20, 30, 40, 50) <class 'tuple'>
(10, 'Python', 3.14, True) <class 'tuple'>
() <class 'tuple'>
() <class 'tuple'>
(99,) <class 'tuple'>
PS C:\Users\tanve\Desktop\ITR>
```

Q2.

PROGRAM:

```
session 7.py > ...
# Tuple Packing
person = ("Shruti", 20, "Pune")

print("Tuple:", person)
print("Type:", type(person))

# Tuple Unpacking
name, age, city = person

print("Name:", name)
print("Age:", age)
print("City:", city)# Tuple Packing
person = ("Shruti", 20, "Pune")

print("Tuple:", person)
print("Type:", type(person))

# Tuple Unpacking
name, age, city = person

print("Name:", name)
print("Age:", age)
print("City:", city)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Users\tanve\Desktop
thon.exe' 'c:\Users\tanve\.vscode\extensions\ms-python.debugpy-
\Users\tanve\Desktop\ITR\session 7.py'
Tuple: ('Shruti', 20, 'Pune')
Type: <class 'tuple'>
Name: Shruti
Age: 20
City: Pune
Tuple: ('Shruti', 20, 'Pune')
Type: <class 'tuple'>
Name: Shruti
Age: 20
City: Pune
PS C:\Users\tanve\Desktop\ITR>
```

Q3.

PROGRAM:

```python
colors = ('red', 'green', 'blue', 'yellow', 'purple', 'orange')

print("First element:", colors[0])
print("Third element:", colors[2])
print("Last element:", colors[-1])
print("Second last element:", colors[-2])

index = int(input("Enter index: "))

if -len(colors) <= index < len(colors):
    print("Element:", colors[index])
else:
    print("Invalid Index")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Users\tanve\Desktop\I
thon.exe' 'c:\Users\tanve\.vscode\extensions\ms-python.debugpy-20
\Users\tanve\Desktop\ITR\session 7.py'
First element: red
Third element: blue
Last element: orange
Second last element: purple
Enter index: 1
Element: green
PS C:\Users\tanve\Desktop\ITR>
```

Q4.

PROGRAM:

```python
session 7.py > ...
nums = (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10)

print("Index 2 to 7:", nums[2:8])
print("First 5 elements:", nums[:5])
print("Last 4 elements:", nums[-4:])
print("Every second element:", nums[::2])
print("Reverse order:", nums[::-1])
```

OUTPUT:

```
thon.exe' 'c:\Users\tanve\.vscode\extensions\ms-python.debug
\Users\tanve\Desktop\ITR\session 7.py'
Index 2 to 7: (2, 3, 4, 5, 6, 7)
First 5 elements: (0, 1, 2, 3, 4)
Last 4 elements: (7, 8, 9, 10)
Every second element: (0, 2, 4, 6, 8, 10)
Reverse order: (10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0)
PS C:\Users\tanve\Desktop\ITR> 
```

Q5.

PROGRAM:

session 7.py > ...

```python
matrix = ((1, 2, 3),
          (4, 5, 6),
          (7, 8, 9))

print("First row:", matrix[0])
print("Second row, third column:", matrix[1][2])
print("Third row, first column:", matrix[2][0])
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Users\ta
thon.exe' 'c:\Users\tanve\.vscode\extensions\ms-pyt
\Users\tanve\Desktop\ITR\session 7.py'
First row: (1, 2, 3)
Second row, third column: 6
Third row, first column: 7
PS C:\Users\tanve\Desktop\ITR>
```

Q6.

PROGRAM:

```python
student = ("shruti", 20, "Computer Science", "A")

# Iteration
for item in student:
    print(item)

# Unpacking
name, age, branch, grade = student

print("\nName:", name)
print("Age:", age)
print("Branch:", branch)
print("Grade:", grade)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Users\ta
 'c:\Users\tanve\.vscode\extensions\ms-python.debug
op\ITR\session 7.py'
shruti
20
Computer Science
A

Name: shruti
Age: 20
Branch: Computer Science
Grade: A
PS C:\Users\tanve\Desktop\ITR>
```

Q7.

PROGRAM:

```
session 7.py > ...
grades = ('A', 'B', 'A', 'C', 'A', 'B', 'D', 'A', 'B')

print("A appears:", grades.count('A'))
print("B appears:", grades.count('B'))

grade = input("Enter grade: ")

print(grade, "appears", grades.count(grade), "times")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Us
  'c:\Users\tanve\.vscode\extensions\ms-python
op\ITR\session 7.py'
A appears: 4
B appears: 3
Enter grade: A
A appears 4 times
PS C:\Users\tanve\Desktop\ITR>
```

Q8.

PROGRAM:

```python
fruits = ('apple', 'banana', 'cherry', 'banana', 'mango', 'apple')

print("First index of banana:", fruits.index('banana'))
print("Banana from index 2:", fruits.index('banana', 2))

try:
    print("Index of kiwi:", fruits.index('kiwi'))
except ValueError:
    print("Not found")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Us
 'c:\Users\tanve\.vscode\extensions\ms-pythor
op\ITR\session 7.py'
First index of banana: 1
Banana from index 2: 3
Not found
PS C:\Users\tanve\Desktop\ITR>
```

Q9.

PROGRAM:

session 7.py > ...

```python
coordinates = (10, 20)

# coordinates[0] = 50    # Error
# coordinates.append(30) # Error

print("Original Tuple:", coordinates)

# Correct way
temp = list(coordinates)
temp[0] = 50
temp.append(30)

coordinates = tuple(temp)

print("Modified Tuple:", coordinates)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd
 'c:\Users\tanve\.vscode\extensions\ms-
op\ITR\session 7.py'
Original Tuple: (10, 20)
Modified Tuple: (50, 20, 30)
PS C:\Users\tanve\Desktop\ITR>
```

Q10.

PROGRAM:

session 7.py > ...

```python
# Input
name = input("Enter Name: ")
roll = int(input("Enter Roll Number: "))
m1 = int(input("Enter Marks 1: "))
m2 = int(input("Enter Marks 2: "))
m3 = int(input("Enter Marks 3: "))

# Packing
student = (name, roll, m1, m2, m3)

print("\nStudent Record:", student)

# Unpacking
name, roll, m1, m2, m3 = student

print("\nName:", name)
print("Roll Number:", roll)
print("Marks 1:", m1)
print("Marks 2:", m2)
print("Marks 3:", m3)

# Count example
mark = int(input("\nEnter mark to count: "))
print("Count:", student.count(mark))

# Nested tuple with 2 students
student2 = ("Amit", 102, 80, 85, 90)

records = (student, student2)

print("\nAll Records:")
for s in records:
    print(s)

print("\nFirst Student Name:", records[0][0])
print("Second Student Roll No:", records[1][1])
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR>  c:; cd 'c:\Users\tanve
 'c:\Users\tanve\.vscode\extensions\ms-python.debugpy-
op\ITR\session 7.py'
Enter Name: shruti
Enter Roll Number: 26
Enter Marks 1: 98
Enter Marks 2: 77
Enter Marks 3: 87

Student Record: ('shruti', 26, 98, 77, 87)

Name: shruti
Roll Number: 26
Marks 1: 98
Marks 2: 77
Marks 3: 87

Enter mark to count: 88
Count: 0

All Records:
('shruti', 26, 98, 77, 87)
('Amit', 102, 80, 85, 90)

First Student Name: shruti
Second Student Roll No: 102
PS C:\Users\tanve\Desktop\ITR>
```